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WHITING SCHOOL
of ENGINEERING

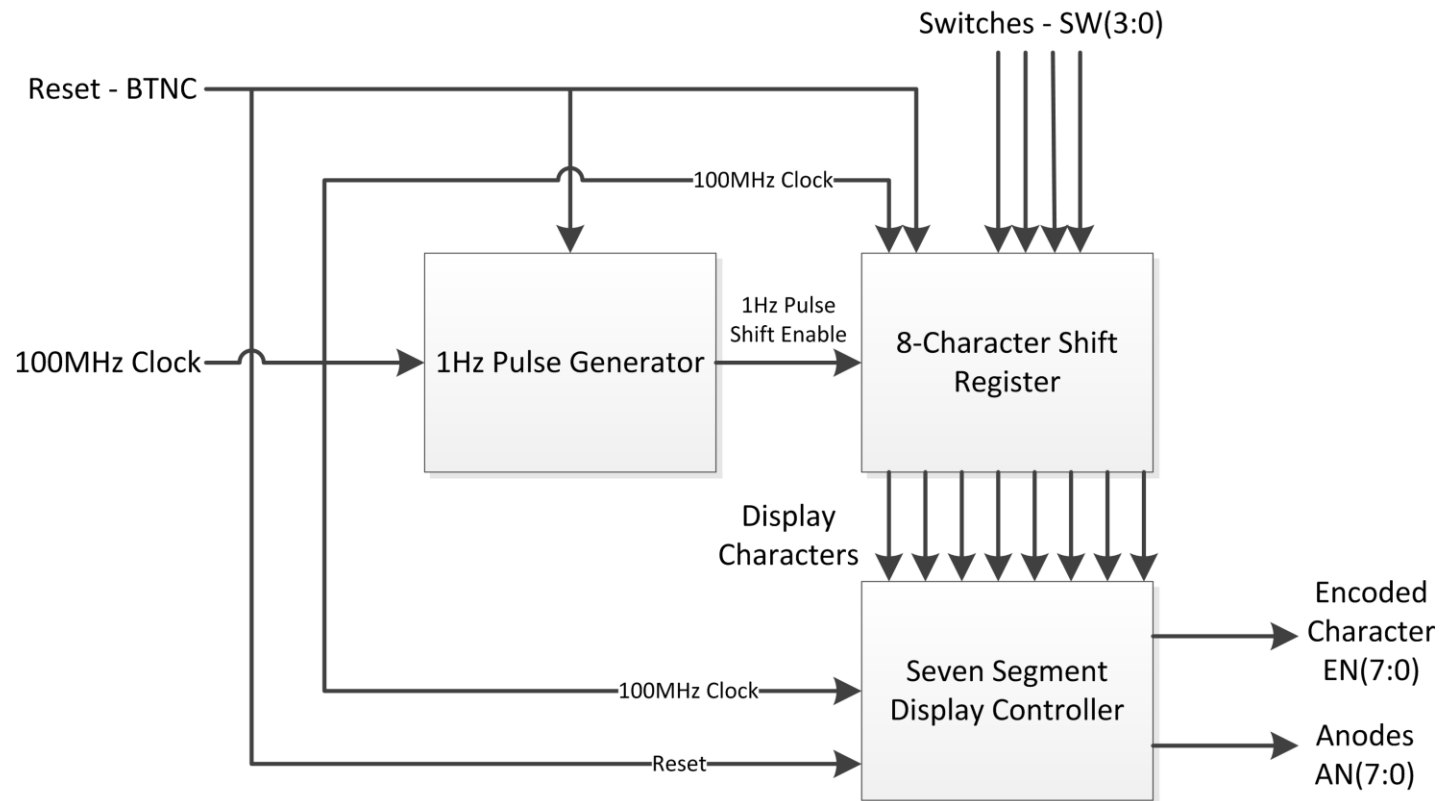
FPGA Design Using VHDL

Module 3E: Lab 3 Overview

Lab 3 Specifications

1. Design a driver that displays **different** digits on the 7-seg displays
 - Use pulse generator to quickly display char0 on display0 then char1 on display1 then char2 on display3, ... to char7 on display7
 - Hint: Look at Nexy's reference manual assigned reading on the 7-seg display and time multiplexing
2. Design a controller that scrolls the digits left at a rate of 1Hz
 - Only start this **AFTER** getting the above to work with fixed values on each display. Ensure you can build a design that shows 12345678 on each display then start working on this task that shifts in a new digit!

Lab 3 Block Diagram



Lab 3: Seven Segment Controller

